

Pseudodeterministic Parity-Query Complexity

Linear Gavinsky Bound - medium

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1 Preliminaries

For $x \in \{0, 1\}^n$, let

$$|x| := |\{i \in [n] : x_i = 1\}|$$

denote its Hamming weight.

Definition 1 (Gap Majority). *The partial Boolean function*

$$\text{GapMaj}_n : \{0, 1\}^n \longrightarrow \{0, 1, *\}$$

is defined by

$$\text{GapMaj}_n(x) := \begin{cases} 0, & |x| \leq n/3, \\ 1, & |x| \geq 2n/3, \\ *, & n/3 < |x| < 2n/3. \end{cases}$$

Definition 2 (Randomized parity-query complexity). *A parity query to an input $x \in \{0, 1\}^n$ asks for the value*

$$\bigoplus_{i \in S} x_i$$

for some set $S \subseteq [n]$. A deterministic parity decision tree is an adaptive algorithm that makes parity queries and outputs a bit.

For a total Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $R_\varepsilon^\oplus(f)$ be the minimum d for which there exists a randomized parity decision tree of worst-case depth at most d that satisfies

$$\Pr[T(x) = f(x)] \geq 1 - \varepsilon$$

for every $x \in \{0, 1\}^n$.

Definition 3 (Pseudodeterministic parity-query complexity). *A completion of a partial Boolean function $g : \{0, 1\}^n \rightarrow \{0, 1, *\}$ is a total Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that $f(x) = g(x)$ whenever $g(x) \neq *$. Define*

$$\text{PsR}_\varepsilon^\oplus(g) := \min\{R_\varepsilon^\oplus(f) : f \text{ is a completion of } g\}.$$

2 Problem formulation

There exists an absolute constant $c > 0$ such that, for all sufficiently large n ,

$$\text{PsR}_{1/3}^{\oplus}(\text{GapMaj}_n) \geq cn.$$

Equivalently, every completion $f : \{0, 1\}^n \rightarrow \{0, 1\}$ of GapMaj_n satisfies

$$\text{R}_{1/3}^{\oplus}(f) \geq cn.$$